

Human Classifier Simulation

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rows Columns and Data Types

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Construct both outcome paths for Constructing a classifier. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Constructing a classifier. Proposed explanation: “Any numeric input must receive a category”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

A written rule selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Recommend a book only if genre is science AND pages $\leq$ 150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

6. Solve the given case

Recommend if adventure AND pages $<$ 150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A human classifier uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what evidence is missing in the second case.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 10 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Recommend a book only if genre is science AND pages $\leq$ 150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Write an input contract

Write the complete Tower1 input contract. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted input does not have to be one of the examples used to fit the threshold.

My reasoning and evidence

13. W2 Construct both outcome paths

Write numbered instructions for Tower1, including input checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

My reasoning and evidence

14. W3 Trace boundaries and a broken path

Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records no prediction for a blank, then continues and treats the blank as zero to report short. Identify the missing instruction and the proper final outcome.

My reasoning and evidence

15. W4 Build an independent ribbon classifier

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise. Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for no prediction and the model version in each record.

My reasoning and evidence

16. W5 Explain what your procedure establishes

Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown. Explain why changing a blank to 1 cm just to produce a category would be a mistake.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Human Classifier Simulation

A fictional librarian wants illustrated mystery books with at most 100 pages.

Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing

Human Classifier Simulation

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

**1. Read all conditions**

Three requirements must pass, not just a high preference score.

↓

**2. Trace J**

Mystery, illustrated and exactly 100 pages: choose.

↓

**3. Trace K and L**

K fails illustration; L fails genre: do not choose.

↓

**4. Trace M**

Missing pages means review rather than assuming it is short.

CHANGE ONE THING
Supply M's pages as 101.

Explain the changed result using the matching lesson.

Why are J and M initially treated differently even though both are illustrated mysteries?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional librarian wants illustrated mystery books with at most 100 pages.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Missing pages means review rather than assuming it is short.

1. Read all conditions
2. Trace M
3. Trace J
4. Trace K and L

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Trace M Missing pages means review rather than assuming it is short. This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

Practice 2: explain the deciding evidence

Changed-case practice: Supply M's pages as 101. Which conclusion is supported by the supplied facts?

1. M is now rejected rather than awaiting page review.
2. We can decide the result without examining the changed input or the procedure.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: M is now rejected rather than awaiting page review. The required field is known but fails the maximum-length condition.

Repair a missing building block

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
